🎮 Robot Handle Control

1. Course Introduction

Control the A1 Pro robot chassis movement and robotic arm joint angles using a handle.

🎯 Course Objectives

1. Master the launch method for the handle control node
2. Familiarize yourself with the functional mapping of handle buttons and joysticks
3. Understand the methods for controlling the chassis and robotic arm with the handle

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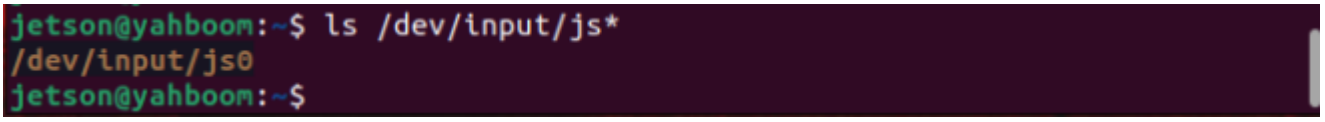
2. Preparation

2.1 Connecting the Handle

Insert the USB handle receiver into the A1 Pro's USB port and check if the handle is recognized by the system:

```
ls /dev/input/js*
```

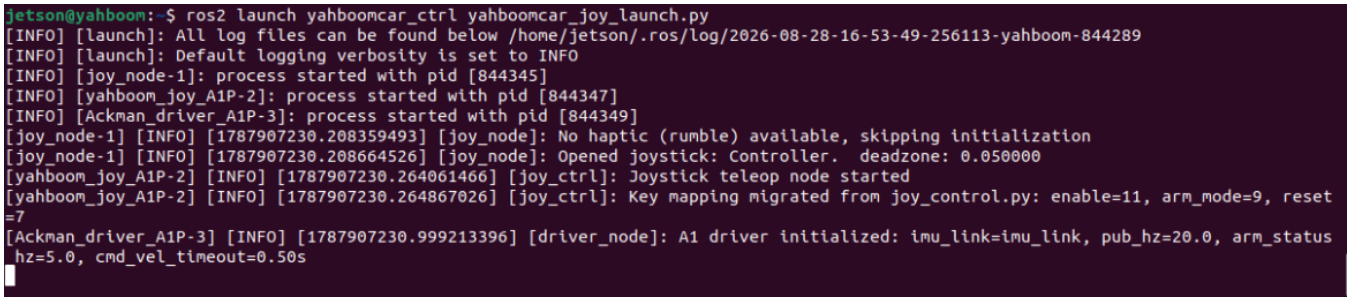
You should normally see the `/dev/input/js0` device.



2.2 Starting Handle Control

Start the chassis driver + handle control with one command:

```
ros2 launch yahboomcar_ctrl yahboomcar_joy_launch.py
```



> **Note**
>
> This launch file will simultaneously start the `joy_node` (handle driver) and `yahboom_joy_A1` (handle control logic) nodes.

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3. Handle Button Operations



3.1 Chassis Motion Control

Input	Function	Description
Left Stick Up/Down (Y-axis)	Forward / Backward	Analog control
Left Stick Left/Right (X-axis)	Turn Left / Right	Analog control

> ⚠ **Warning**
>
> The A1 Pro uses an Ackermann steering structure. When turning, you need to simultaneously push the forward/backward stick; turning left/right alone will not move the vehicle.

3.2 Speed Adjustment

Button	Function
Button L1	Toggle linear velocity gear
Button L2	Toggle angular velocity gear

3.3 Robotic Arm Control

The handle supports two robotic arm control modes, switched via **Button 9**:

Mode A (Default):

Input	Function
D-pad Left/Right	Joint 1 rotation
D-pad Up/Down	Joint 2 rotation

Mode B:

Input	Function
D-pad Left/Right	Joint 5 rotation
D-pad Up/Down	Joint 6 rotation

Universal Buttons:

Button	Function
Button X	Joint 3 increase
Button B	Joint 3 decrease
Button Y	Joint 4 increase
Button A	Joint 4 decrease

Button	Function
Button SELECT	Reset robotic arm to initial pose

3.4 Enable/Disable Switch

Button	Function
Button (START)	Enable/disable handle control

> **Caution**
>
> The handle is disabled by default; on first use, press START to activate control.

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4. Source Code Reference

Source code path:

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_ctrl/yahboomcar_ctrl/yahboom_joy_A1P.py
#rdk
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_ctrl/yahboomcar_ctrl/yahboom_joy_A1p.py
```

The handle node subscribes to the `/joy` topic to receive button and joystick states, then publishes to `/cmd_vel` (chassis) and `/arm_joint` (robotic arm) topics after dead zone filtering and velocity mapping.